



Modelling Financial Markets

Seminar 17: Bubbles and Crashes — Hands-on



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BIC: Bayesian Information Criterion
- ▣ BSADF: Backward Sup Augmented Dickey–Fuller (statistic)
- ▣ CI: Confidence Interval
- ▣ EDGAR: Electronic Data Gathering, Analysis, and Retrieval (SEC filings database)
- ▣ EM: Expectation-Maximisation (algorithm)
- ▣ FCLT: Functional Central Limit Theorem
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ JLS: Johansen–Ledoit–Sornette (model)
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ LR: Likelihood Ratio (test)
- ▣ MC: Monte Carlo (simulation)
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ OLS: Ordinary Least Squares
- ▣ PSY: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- ▣ PWY: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SEC: Securities and Exchange Commission
- ▣ US: United States

Today's Question and Required Outputs

- Question of the seminar: **did a price rise faster than a random walk allows, since when, and how sure can we be?**
- Route
 - ▶ A1, A3, A5 on paper: the rational bubble, the right-tailed critical value, one step of the regime filter
 - ▶ B1, B3: explosive episodes in the price-dividend ratio and in Bitcoin, with simulated and bootstrap critical values
 - ▶ B5: how far an LPPLS critical time moves with the window; B7: two regimes for the S&P 500
 - ▶ C2: find the errors in an AI answer; B4 at home
- Required outputs (checklist on the last core slide): the A answers, one figure and one table per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Background: Lecture 17; textbook [Franke, Härdle & Hafner \(2019\)](#)
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	can a price exceed the value of the dividends, and how does the gap grow?	Solved	–
A3	why is the right-tailed critical value negative?	Solved	–
A5	how much does one bad week move the regime probability?	Solved	–
B1	is the S&P 500 price-dividend ratio explosive, and when?	Solved	A3
B3	does changing volatility change Bitcoin's episodes?	Solved	B1
B5	how far does the LPPLS critical time move with the window?	Solved	–
B7	when was the S&P 500 in its turbulent regime?	Solved	A5
C2	what is wrong in an AI answer on SADF and LPPLS?	Proposed	A3, B1, B5
B4	is the Bucharest Stock Exchange explosive today?	Proposed	B1, B3
Extension packs	A2, A4, A6–A8, B2, B6, B8–B10, C1, C3 (optional, after the seminar)	both	named on each task

- ☐ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ☐ Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Manifest

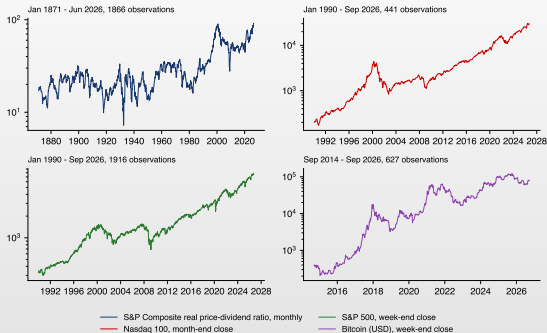
Series	Source, field	Frequency, sample	Used in
S&P Composite, real price and dividend	Robert J. Shiller, shillerdata.com	monthly, Jan 1871–Jun 2026	B1, C2
S&P 500	GSPC.INDX, close	weekly and monthly, 1990–2026	B7, B2
Nasdaq 100	NDX.INDX, close	monthly, daily, 1990–2026	B2, B6, C1
Bitcoin	BTC-USD.CC, close	weekly, daily, 2014–2026	B3, B5, B8
BET, BET-FI	BET, BETFI.INDX, close	monthly, weekly, 1997–2026	B4
NVIDIA, Cisco	NVDA.US, CSCO.US, adjusted_close	monthly	C1
US house prices and new-home sales	FRED CSUSHPINSA, HSN1FNBSA	monthly, 1990–2011	C3

- Market files: `data/market/<SYMBOL>.csv`, columns `date`, `open`, `high`, `low`, `close`, `adjusted_close`, `volume`; closes up to 18 September 2026
- Weekly and monthly series: last close of the week or month; tests use log levels $y_t = \ln P_t$ (or $\ln(P_t/D_t)$)

Setup: Run and Verify

```
# notebook section "Data" defines
# price(name, freq, start) and shiller()
pd_ratio = shiller()['PD']           # monthly
ndx = price('ndx', 'M')              # month-end
spx = price('sp500', 'W', '1990-01-01')
btc = price('btc', 'W')              # week-end
for x in (pd_ratio, ndx, spx, btc):
    print(len(x), x.index[0].date(),
          x.index[-1].date())
```

- Expected: 1866 (P/D, Jan 1871–Jun 2026), 441 (Nasdaq 100), 1916 (S&P 500), 627 (Bitcoin)
- If a count differs, stop and check the name, the frequency and the start date



Notation Used in Every Exercise

- y_t : log price or log price-dividend ratio; right-tailed ADF regression $\Delta y_t = a + b y_{t-1} + e_t$ with a constant and no lags; $t_b = \hat{b}/SE(\hat{b})$; $H_0 : b = 0$ vs $H_1 : b > 0$
- T : number of changes; minimum window $w_0 = \lfloor r_0 T \rfloor$, $r_0 = 0.01 + 1.8/\sqrt{T}$; $ADF'_{r_1}^2$: the statistic on the window from fraction r_1 to r_2 of the sample
- $SADF = \sup_{r_2} ADF_0'^2$; $BSADF_{r_2} = \sup_{r_1} ADF_{r_1}'^2$; $GSADF = \sup_{r_2} BSADF_{r_2}$ [Phillips, Shi & Yu \(2015a\)](#)
- **Episode**: BSADF above its pointwise 95% critical value for at least $L = \lceil \ln T \rceil$ consecutive periods; start dated in hindsight, **confirmed** $L - 1$ periods later; **rise** or **fall**: sign of the price change over the episode
- Markov switching: regime s_t , $p_{ij} = P(s_t = j \mid s_{t-1} = i)$, $\varphi(x; \mu, \sigma^2)$ the Normal density; **filtered** probability: data up to t ; **smoothed**: the whole sample
- LPPLS: $\ln p_t = A + B(t_c - t)^m + (t_c - t)^m [C_1 \cos(\omega \ln(t_c - t)) + C_2 \sin(\omega \ln(t_c - t))]$; a **qualified** fit passes the filter of Shu and Zhu (eq. 12) [Shu & Zhu \(2020\)](#)

A1 [Solved]: Present Value and the Rational Bubble — Task

- **Question:** can a price exceed the present value of its dividends in a rational market, and what happens to the gap over time?
- **Inputs:** a share with price $P_t \geq 0$ and dividends $D_t \geq 0$ whose discounted expected sum is finite [Blanchard & Watson \(1982\)](#)
 - ▶ Model assumption, a constant required return $r > 0$: $P_t = \mathbb{E}_t[P_{t+1} + D_{t+1}]/(1 + r)$
 - ▶ Numbers for task 4: $D_1 = 2$, $g = 2\%$, $r = 6\%$, $P = 65$
- **Tasks**
 1. Substitute the pricing equation into itself h times and write P_t as discounted expected dividends up to $t + h$ plus a remainder.
 2. Let $h \rightarrow \infty$, define F_t (the fundamental value) and $B_t = P_t - F_t$, and prove that $\mathbb{E}_t[B_{t+1}] = (1 + r)B_t$.
 3. Prove that $B_t \geq 0$ and that $B_t = 0$ implies $B_{t+k} = 0$ for every $k \geq 1$ [Diba & Grossman \(1988\)](#).
 4. In the Gordon case compute $F = D_1/(r - g)$, $B = P - F$ and the bubble share $\mathbb{E}[B_h]/(F_h + \mathbb{E}[B_h])$ for $h = 5, 30, 100$ years.
- **Report:** the derivation, F , B and the three shares
- **Interpretation:** why must a rational bubble exist from the first trade, and why is a growing bubble share the testable implication?
- **Notebook:** section A1

A1 [Solved]: Solution (1/2) — Steps 1–3, the Derivation

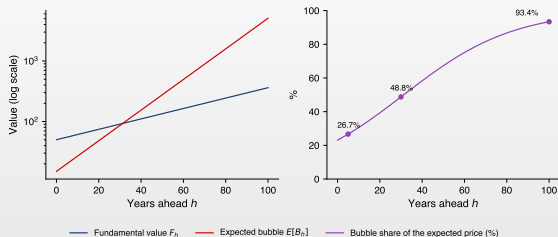
- Step 1: write $P_{t+1} = \mathbb{E}_{t+1}[P_{t+2} + D_{t+2}]/(1+r)$ and substitute; iterated expectations give $\mathbb{E}_t \mathbb{E}_{t+1} = \mathbb{E}_t$; after h steps

$$P_t = \sum_{i=1}^h \frac{\mathbb{E}_t D_{t+i}}{(1+r)^i} + \frac{\mathbb{E}_t P_{t+h}}{(1+r)^h}$$
- Step 2: with $D \geq 0$ the partial sums increase to the finite $F_t = \sum_{i \geq 1} \mathbb{E}_t D_{t+i}/(1+r)^i$; the remainder converges to

$$B_t := P_t - F_t = \lim_h \mathbb{E}_t P_{t+h}/(1+r)^h$$
 - ▶ $(1+r)F_t = \mathbb{E}_t D_{t+1} + \mathbb{E}_t F_{t+1}$ (iterated expectations, monotone convergence); subtract it from $(1+r)P_t = \mathbb{E}_t D_{t+1} + \mathbb{E}_t P_{t+1}$: $\mathbb{E}_t B_{t+1} = (1+r)B_t$
 - ▶ Transversality, $\lim_h \mathbb{E}_t P_{t+h}/(1+r)^h = 0$, is exactly $B_t = 0$
- Step 3: B_t is the limit of the remainders $\mathbb{E}_t P_{t+h}/(1+r)^h \geq 0$, so $B_t \geq 0$; if $B_t = 0$, then $\mathbb{E}_t B_{t+1} = 0$ with $B_{t+1} \geq 0$, so $B_{t+1} = 0$ almost surely, and by induction $B_{t+k} = 0$

A1 [Solved]: Solution (2/2) — Step 4, Numbers and Chart

- Step 4: $F = 2/(0.06 - 0.02) = 50.0$;
 $B = 65 - 50.0 = 15.0$, i.e. 23.1% of the price
- $F_h = 50.0 \cdot 1.02^h$, $\mathbb{E}B_h = 15.0 \cdot 1.06^h$
- Bubble share: 26.7% ($h = 5$), 48.8% ($h = 30$), 93.4% ($h = 100$)
- Interpretation: a bubble cannot start from zero later; since $r > g$ its **expected** value eventually dominates the price, so the expected price path is explosive relative to fundamentals: this is what right-tailed tests look for



A3 [Solved]: The Right-Tailed Dickey–Fuller Distribution — Task

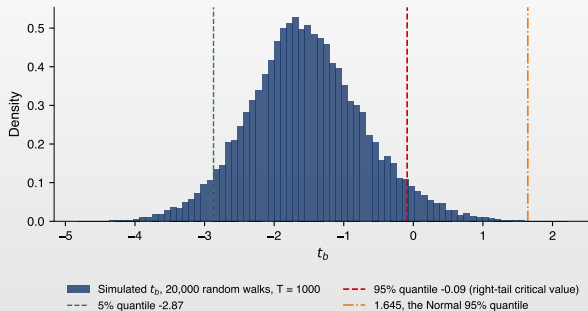
- **Question:** under a unit root, where does the right-tailed ADF statistic lie, and why is 1.645 the wrong critical value?
- **Inputs:** $y_t = y_{t-1} + \varepsilon_t$, ε_t iid $(0, \sigma^2)$, $y_0 = 0$; regression $\Delta y_t = a + b y_{t-1} + e_t$, $t = 1, \dots, T$
 - ▶ FCLT: $T^{-1/2} y_{\lfloor Tr \rfloor} \Rightarrow \sigma W(r)$, with W a standard Brownian motion; $\tilde{y}_{t-1}$: y_{t-1} minus its sample mean
- **Tasks**
 1. Write $\hat{b}$ with the demeaned regressor $\tilde{y}_{t-1}$.
 2. Using the FCLT and continuous mapping, find the limits of $T^{-2} \sum \tilde{y}_{t-1}^2$ and $T^{-1} \sum \tilde{y}_{t-1} \varepsilon_t$.
 3. Derive the limit of t_b and show that the limit numerator has mean $-1/2$.
 4. Simulate 20 000 random walks with $T = 1\,000$ and report the 5%, 50% and 95% quantiles of t_b and the share of negative values.
- **Report:** the limit of t_b , the three quantiles, the share of negative values and a histogram
- **Interpretation:** what goes wrong if a bubble test uses 1.645?
- **Notebook:** section A3

A3 [Solved]: Solution (1/2) — Steps 1–3, the Limit

- Step 1: OLS with a constant, $\hat{b} = \sum \tilde{y}_{t-1} \Delta y_t / \sum \tilde{y}_{t-1}^2 = \sum \tilde{y}_{t-1} \varepsilon_t / \sum \tilde{y}_{t-1}^2$ under H_0
- Step 2: $T^{-2} \sum \tilde{y}_{t-1}^2 \Rightarrow \sigma^2 \int_0^1 \tilde{W}^2$, $\tilde{W} = W - \int_0^1 W$; $T^{-1} \sum y_{t-1} \varepsilon_t \Rightarrow \frac{\sigma^2}{2} (W(1)^2 - 1)$, and demeaning subtracts $\sigma^2 W(1) \int_0^1 W$
- Step 3: $\text{SE}(\hat{b}) = s / \sqrt{\sum \tilde{y}_{t-1}^2}$ with $s \rightarrow \sigma$, so σ cancels (PSY, eq. 5 with $r_1 = 0$, $r_2 = 1$) [Phillips, Shi & Yu \(2015a\)](#)
 - ▶ $t_b \Rightarrow \frac{\frac{1}{2}(W(1)^2 - 1) - W(1) \int_0^1 W}{\{\int_0^1 W^2 - (\int_0^1 W)^2\}^{1/2}}$
 - ▶ $\mathbb{E}[W(1)^2] = 1$ and $\mathbb{E}[W(1) \int_0^1 W] = \int_0^1 r dr = \frac{1}{2}$: the numerator has mean $\frac{1}{2}(1 - 1) - \frac{1}{2} = -\frac{1}{2}$ (simulated: -0.50)

A3 [Solved]: Solution (2/2) — Step 4, Simulation

- Step 4: 20 000 walks, $T = 1\,000$; quantiles 5%: -2.87 , 50%: -1.57 , 95%: -0.09
- 96% of the t_b are negative: the whole distribution lies left of zero
- Interpretation: the estimated constant absorbs part of the random walk and pushes $\hat{b}$ down; with 1.645 a right-tailed test almost never rejects: its size is close to zero and its power very low



A5 [Solved]: Regime Durations and One Filter Step — Task

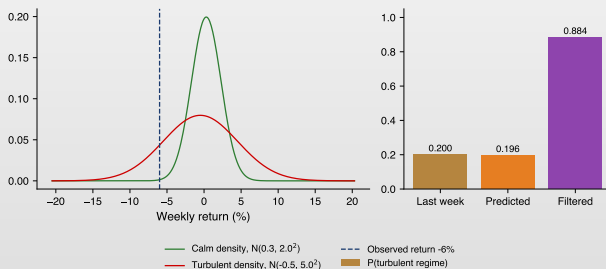
- **Question:** how much does one bad week move the probability of the turbulent regime?
- **Inputs:** weekly returns in %, $r_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t$, $\varepsilon_t \sim N(0, 1)$ [Hamilton \(1989\)](#)
 - ▶ Calm: $\mu_1 = 0.3$, $\sigma_1 = 2$; turbulent: $\mu_2 = -0.5$, $\sigma_2 = 5$; $p_{11} = 0.98$, $p_{22} = 0.90$
 - ▶ Last week $P(s = 2 \mid \text{data}) = 0.2$; this week $r_t = -6\%$
- **Tasks**
 1. Compute the expected duration $1/(1 - p_{ii})$ of each regime.
 2. Compute the stationary probabilities $\pi_2 = p_{12}/(p_{12} + p_{21})$ and $\pi_1 = 1 - \pi_2$.
 3. Compute the unconditional mean and standard deviation of r_t with the law of total variance.
 4. Do one Hamilton filter step: the predicted probability, the two densities at $r_t = -6$, the updated probability.
- **Report:** the durations, π_1 , π_2 , mean and standard deviation, and the three probabilities of step 4
- **Interpretation:** why does one week change the regime probability so much?
- **Notebook:** section A5

A5 [Solved]: Solution (1/2) — Steps 1–3

- Step 1: $1/(1 - 0.98) = 50$ weeks (calm), $1/(1 - 0.90) = 10$ weeks (turbulent)
- Step 2: $\pi_2 = 0.02/(0.02 + 0.10) = 0.167$, $\pi_1 = 0.833$
- Step 3: $\mathbb{E}r = 0.833 \cdot 0.3 + 0.167 \cdot (-0.5) = 0.167$; $\text{Var}(r) = \sum_i \pi_i \sigma_i^2 + \sum_i \pi_i (\mu_i - \mathbb{E}r)^2 = 7.59$, s.d. 2.75

A5 [Solved]: Solution (2/2) — Step 4, the Filter Step

- Step 4: prediction $0.2 \cdot 0.90 + 0.8 \cdot 0.02 = 0.196$
 - ▶ Densities $\varphi(-6; 0.3, 2^2) = 0.00140$, $\varphi(-6; -0.5, 5^2) = 0.04357$
 - ▶ Update $\frac{0.196 \cdot 0.04357}{0.196 \cdot 0.04357 + (1 - 0.196) \cdot 0.00140} = 0.884$
- Interpretation: -6% lies more than three calm standard deviations below the calm mean, so its calm density is tiny: one such week almost settles the regime



B1 [Solved]: Is the S&P 500 Price-Dividend Ratio Explosive? — Task

- **Question:** is the S&P 500 price-dividend ratio explosive, when, and does the answer depend on the minimum window?
- **Model:** A3
- **Data:** S&P Composite real price and real dividend, monthly 1871–2026; $y_t = \ln(P_t/D_t)$, $T = 1\,865$ changes
- **Tasks**
 1. Compute GSADF (constant, no lags) with the minimum window $r_0 = 0.01 + 1.8/\sqrt{T}$.
 2. Simulate 1000 random walks $y_t = T^{-1} + y_{t-1} + \varepsilon_t$ and take the 95% quantile of GSADF and, date by date, of BSADF [Phillips, Shi & Yu \(2015a\)](#).
 3. Repeat steps 1–2 with $r_0 = 0.03$ and $r_0 = 0.08$.
 4. For each r_0 , date-stamp the episodes (at least $L = \lceil \ln T \rceil$ months above the critical value) and mark each as a rise or a fall of the ratio.
- **Report:** a table (r_0 , w_0 , GSADF, critical value, episodes) and a chart of BSADF against its critical values
- **Interpretation:** which episodes survive every window, and is every explosive episode of the ratio a bubble?
- **Notebook:** section B1

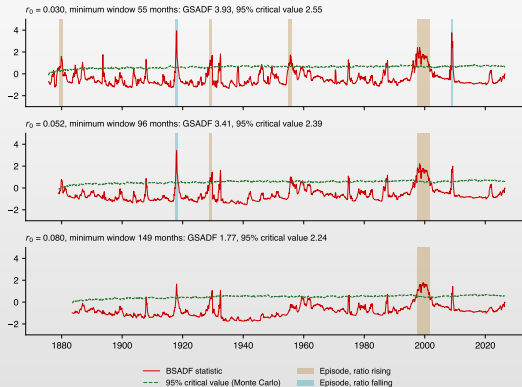
B1 [Solved]: Solution (1/2) — Steps and Table

- Steps 1–2: log ratio; for every end date, ADF on all windows of at least w_0 months; the supremum over start dates is BSADF, over end dates GSADF; the same computation on the simulated walks gives the critical values
- Steps 3–4, results (episodes that are a **fall** of the ratio are marked):

r_0	w_0	GSADF	cv 95%	Episodes (BSADF)
0.030	55	3.93	2.55	Jan 1879–Apr 1880; Aug 1917–May 1918 (falling); Nov 1928–Sep 1929; Dec 1954–Jan 1956; May 1997–Aug 2001; Sep 2008–Apr 2009 (falling)
0.052	96	3.41	2.39	Aug 1917–Jun 1918 (falling); Nov 1928–Sep 1929; May 1997–Aug 2001
0.080	149	1.77	2.24	May 1997–Aug 2001

- The dot-com episode (1997–2001) is dated with every window; with $r_0 = 0.08$ GSADF is below its critical value, yet BSADF still crosses: the global test is stricter than the date-stamping
- The shortest window adds Jan 1879–Apr 1880; Dec 1954–Jan 1956; Sep 2008–Apr 2009 (falling): fewer observations per regression, noisier ADF statistics, more crossings
- 1917–1918 is a **fall** of the ratio (real prices dropped faster than real dividends): explosive, but not a bubble
- Interpretation: only episodes that survive every r_0 and point upwards support the bubble reading

B1 [Solved]: Solution (2/2) — Chart

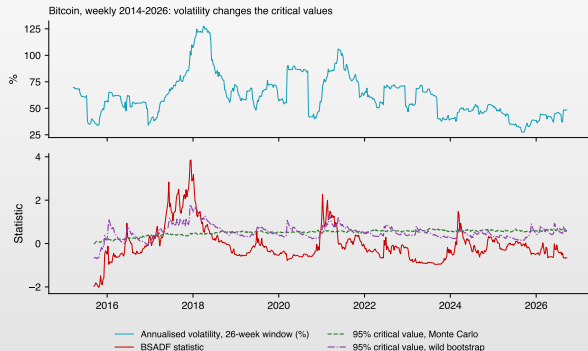


Red: BSADF; green: its pointwise 95% critical value; shaded: date-stamped episodes, rising (sand) or falling (blue) ratio

B3 [Solved]: Bitcoin: Simulated vs Bootstrap Critical Values — Task

- ▣ **Question:** does Bitcoin's changing volatility change the verdict of the test and the dates of the episodes?
- ▣ **Model:** B1
- ▣ **Data:** weekly log closes of Bitcoin, September 2014 – September 2026, $T = 626$ weekly changes
- ▣ **Tasks**
 1. Compute GSADF with the minimum-window rule and Monte Carlo critical values from 2 000 Gaussian random walks.
 2. Wild bootstrap: multiply the demeaned weekly changes by independent random signs $w_t = \pm 1$, rebuild the path and recompute BSADF and GSADF, 1 000 times; take the 95% quantiles [Harvey et al. \(2016\)](#).
 3. Date-stamp the episodes with both critical-value sequences, minimum duration $\lceil \ln T \rceil$ weeks.
 4. Plot BSADF with both sequences, and the 26-week rolling volatility above it.
- ▣ **Report:** GSADF, both 95% critical values, both lists of episodes and the chart
- ▣ **Interpretation:** when are the bootstrap critical values higher, and which episode is robust to the choice?
- ▣ **Notebook:** section B3

B3 [Solved]: Solution

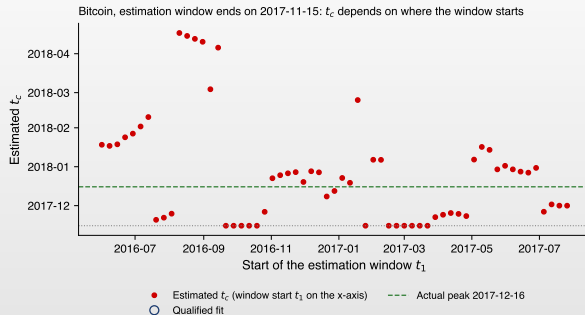


- Steps 1–2: $GSADF = 3.85$; 95% critical value: Monte Carlo 2.27, wild bootstrap 2.82; $3.85 > 2.82$: the random walk is rejected at 5% with either
- Step 3, Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
 - Wild bootstrap: May 2017–Jan 2018; Dec 2020–Feb 2021
- Step 4: the bootstrap critical values rise in and after the volatile periods (maximum 1.80 vs 0.71), because the resampled changes keep the large shocks where they occurred
- Interpretation: the 2017 episode survives both, shorter with the bootstrap; the 2021 episode shrinks: its explosiveness is partly a volatility effect

B5 [Solved]: How Much Does the Critical Time Depend on the Window? — Task

- **Question:** how far does the LPPLS critical time t_c move when only the start of the estimation window changes?
- **Data:** Bitcoin daily log closes; 61 windows, all ending on $t_2 = 15$ November 2017, starting every 7 days from 1 June 2016 to 1 August 2017
 - ▶ **Edge fit:** m or ω at the boundary of the search space ($m \leq 0.002$ or ≥ 0.998 ; $\omega \leq 1.01$ or ≥ 49.9)
- **Tasks**
 1. Fit LPPLS on each window: (A, B, C_1, C_2) by OLS, (t_c, m, ω) by a grid and Nelder–Mead in the search space of Shu and Zhu (eq. 11) [Filimonov & Sornette \(2013\)](#), [Shu & Zhu \(2020\)](#).
 2. Report the minimum, median and maximum t_c and the interquartile range of t_c minus the actual peak.
 3. Report the shares of qualified fits (eq. 12), of edge fits and of fits within 30 days of the peak.
 4. Plot t_c against the start of the window, with the actual peak.
- **Report:** the three dates, the interquartile range, the three shares and the chart
- **Interpretation:** would you trust a single t_c reported without its window?
- **Notebook:** section B5

B5 [Solved]: Solution



- Steps 1–2: t_c ranges from 15 Nov 2017 to 17 Apr 2018, median 23 Dec 2017; actual peak 16 Dec 2017; interquartile range of the error 53 days, median error 7 days
- Step 3: qualified fits 0%; edge fits 43%; within 30 days of the peak 56%
- An edge fit is not an interior optimum: the data do not pin the parameter down
- Interpretation: the same end date gives critical times months apart, depending only on the start of the window; a single t_c is not a forecast

B7 [Solved]: Markov Switching on the S&P 500 — Task

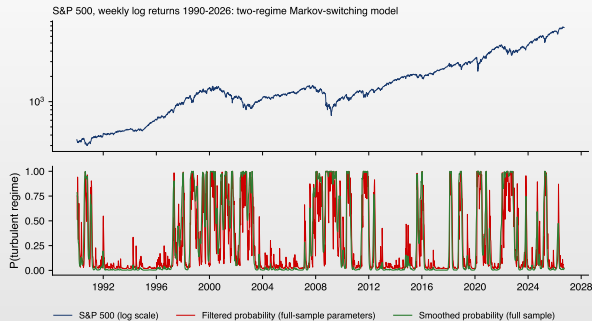
- ▣ **Question:** when was the S&P 500 in its turbulent regime, and is the turbulent mean negative?
- ▣ **Model:** A5
- ▣ **Data:** weekly log returns of the S&P 500 in %, 1990–2026 (1 915 weeks)
- ▣ **Tasks**
 1. Estimate $r_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t$ with two regimes by maximum likelihood (Hamilton filter, 20 random starts); report estimates, standard errors and expected durations [Hamilton \(1989\)](#).
 2. Find the first week of 2000, of autumn 2008 and of 2020 with a filtered turbulent probability above 0.5.
 3. Compute the share of weeks in which the filtered and the smoothed regime (probability above 0.5) agree.
 4. Test $\mu_{\text{turb}} = 0$ with a t -test at 5% and compute $\text{LR} = 2(\hat{\ell}_2 - \hat{\ell}_1)$ against one regime with Normal returns.
- ▣ **Report:** the estimates table, the three dates, the agreement share, t , LR and a chart of the regime probabilities
- ▣ **Interpretation:** is the turbulent mean significantly negative, and why can LR not be compared with a χ^2 critical value?
- ▣ **Notebook:** section B7

B7 [Solved]: Solution (1/2) — Steps 1–3

Regime	$\hat{\mu}$	SE	$\hat{\sigma}$	$\hat{p}_{ii}$	Duration (weeks)
Calm	0.32	0.04	1.49	0.974	38.1
Turbulent	-0.27	0.17	3.67	0.931	14.4

- Step 1: the table; the calm regime covers 73% of the weeks; weekly volatility 3.67% in the turbulent regime vs 1.49% in the calm one
- Step 2: first week above 0.5: 28 Jan 2000 (2000), 3 Oct 2008 (autumn 2008), 28 Feb 2020 (2020)
- Step 3: filtered and smoothed regimes agree in 90% of weeks; the filtered probabilities use full-sample parameters, so they are pseudo real time

B7 [Solved]: Solution (2/2) — Step 4 and Chart



- Step 4: $t = -0.27/0.17$, $|t| < 1.96$: do not reject $\mu_{\text{turb}} = 0$ at 5%
- LR = $2(-4070.0 - (-4326.0)) = 512.1$, but under one regime the transition probabilities are not identified, so χ^2 does not apply; a parametric bootstrap does (B8)
- Interpretation: the turbulent regime marks the stress periods of 2000, 2008 and 2020, but its mean is not significantly negative: regimes describe risk, not predictable losses

C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: “y is the monthly log price-dividend ratio of the S&P 500. Write Python code for SADP with the PSY minimum window, decide at 5% whether there is a bubble, date-stamp it and tell me when it will burst.”
- **Answer:** the code below and the conclusion

```
import numpy as np
from statsmodels.tsa.stattools import adfuller
T = len(y)
w0 = int(np.floor((0.01 + 1.8 / np.sqrt(T)) * T))
adf = lambda x: adfuller(x, maxlag=0, regression='c', autolag=None)[0]
sadf = max(adf(y[:t]) for t in range(w0, T + 1))
cv = adfuller(y, maxlag=0, regression='c', autolag=None)[4]['5%']
print('bubble' if sadf > cv else 'no_bubble')
flag = [adf(y[t - w0:t + w0]) > cv for t in range(w0, T - w0)]
```

- **Conclusion of the AI:** “An LPPLS fit on the last 36 months gives $\hat{t}_c = \text{March 2027}$: this is when the bubble bursts.”

C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Model:** A3, B1 and B5 [Solved]
- ▣ **Data:** the log price-dividend ratio of B1 and simulated random walks of the same length
- ▣ **Tasks**
 1. Find the three errors in the answer: the line and why it is wrong.
 2. For each error, say how you would detect it without knowing the answer: a test on simulated data, a truncated sample with the same design and random seed, a second window.
 3. Write the corrected code: simulated critical values, BSADF on windows that end at t , episodes of at least $\lceil \ln T \rceil$ months.
 4. Log each error you actually found in the format of [AI_ERRORS.md](#): request, answer, error, how found, correction.
- ▣ **Interpretation:** after the corrections, what can the code say about March 2027?
- ▣ AI code usually runs without an error message: the mistakes are in the statistics, not in the syntax

B4 [Proposed]: The Bucharest Stock Exchange — Task

- **Question:** was the 2003–2007 boom explosive in real time, and is the current rise of the BET and BET-FI explosive?
- **Model:** B1 and B3 [Solved]
- **Data:** official BET index, monthly log closes 1997–2026; BET-FI (financial investment companies), weekly log closes 2012–2026
- **Tasks**
 1. For each index compute GSADF with Monte Carlo (2 000 paths) and wild bootstrap (1 000 replications) 95% critical values.
 2. Date-stamp the episodes with both critical-value sequences (minimum duration $\lceil \ln T \rceil$) and classify each as a rise or a fall.
 3. Plot BSADF with both sequences for the two indices.
 4. Find the confirmation date of the 2003–2007 episode (the L -th consecutive exceedance).
- **Report:** a table (T , GSADF, both critical values), the episodes with their direction and the chart
- **Interpretation:** does an explosive current rise make it a bubble (herding and log-periodic patterns before 2007: [Pele, Mazurencu & Nijkamp \(2013\)](#))?
- **Notebook:** section B4

Submission Checklist and Team Repository

Hand in before the next seminar

- ▣ A1, A3, A5: the derivations and the numbers
- ▣ B1, B3, B5, B7: one figure and one table each, as in the solutions
- ▣ C2: corrected code, the three errors, your error log
- ▣ B4: table, episodes and chart
- ▣ One interpretation sentence per exercise

Team repository

- ▣ Teams of 3–4; one public GitHub repository, instructors as collaborators
- ▣ Quantlet folder for B4 (code, Metainfo, charts) with one more asset of your choice (a stock listed in Bucharest, a crypto-asset or a sector index)
- ▣ Log AI use in `AI_USE.md` and caught errors in `AI_ERRORS.md`
- ▣ Extension packs are optional; C1 and C3 are possible project starts

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Rational bubbles and limit theory	A2, A4	A1, A3
2. Regimes	A6, B8	A5, B7
3. LPPLS	A7 [Solved], A8, B6	B5, B3; the lecture section on LPPLS
4. Size under changing volatility	B9 [Solved], B10	B3
5. Choice of windows	B2	B1
6. Project starts	C1, C3	B1–B7, B6; the lecture case study

- Each pack has its own notebook section; within a pack, do the exercises in order
- Solutions of the [Proposed] exercises are discussed in class



A2 [Proposed]: The Blanchard–Watson Bubble — Task

- ▣ **Question:** who pays for the crash risk of a bursting bubble, and why does a full-sample test miss it?
- ▣ **Model:** A1 [Solved]
- ▣ **Inputs:** $B_{t+1} = \frac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π , $B_{t+1} = \varepsilon_{t+1}$ with probability $1 - \pi$, independent of the past; $\mathbb{E}\varepsilon = 0$; $B_0 = 15$, $r = 6\%$, $\pi = 0.9$ per year [Blanchard & Watson \(1982\)](#)
 - ▶ Evans form (positive restart): collapse to $B_{t+1} = \delta$, $0 < \delta \leq B_0$; survival $B_{t+1} = \delta + \frac{1+r}{\pi}(B_t - \frac{\delta}{1+r})$; $y = \ln B$, $c = \ln \delta$
- ▣ **Tasks**
 1. Derive $\mathbb{E}_t B_{t+1}$ and compare it with the martingale condition of A1.
 2. For $\varepsilon \equiv 0$, derive the conditional standard deviation $\text{sd}_t(B_{t+1})$ and evaluate it at $B_t = 15$.
 3. Give the distribution of the lifetime N , then $\mathbb{E}N$, $P(N > 5)$ and the growth rate while the bubble survives.
 4. For the Evans form, check $\mathbb{E}_t B_{t+1} = (1+r)B_t$ and derive $\mathbb{E}[\Delta y_{t+1} \mid y_t]$ for $B_t \gg \delta$ and its slope in y_t .
- ▣ **Report:** the four derivations with the numbers
- ▣ **Interpretation:** what does a full-sample ADF regression on $\ln B$ see, and why is this the Evans pitfall?
- ▣ **Notebook:** section A2

A4 [Proposed]: A Mildly Explosive Root — Task

- **Question:** is the right-tailed test consistent against a bubble whose root is only slightly above one, and how fast does its statistic grow?
- **Model:** A3 [Solved]
- **Inputs:** $y_t = \rho_T y_{t-1} + \varepsilon_t$, $y_0 = 0$, ε_t iid $(0, \sigma^2)$, $\rho_T = 1 + c/k_T$, $c > 0$, $k_T \rightarrow \infty$, $k_T = o(T)$ Phillips & Magdalinos (2007)
- **Tasks**
 1. Write $y_t = \rho_T^t X_t$ with $X_t = \sum_{j \leq t} \rho_T^{-j} \varepsilon_j$ and find the limit of $\text{Var}(X_T)(\rho_T^2 - 1)$.
 2. Show that $\sum y_{t-1}^2 \approx X_T^2 \rho_T^{2T} / (\rho_T^2 - 1)$ and that $\frac{\rho_T^T}{\rho_T^2 - 1}(\hat{\rho} - \rho_T)$ is asymptotically a ratio of two independent Normal variables.
 3. Decompose the t -ratio of $b = \rho - 1$ and derive its divergence rate.
 4. Simulate 4000 paths with $c = 1$, $k_T = T^{0.8}$, $T = 100, 400, 1600$ (regression with a constant); report ρ_T^T , the median t -ratio and a histogram.
- **Report:** the three derivations and the simulation table with the histogram
- **Interpretation:** what does the divergence rate say about the power against short, mild bubbles?
- **Notebook:** section A4

A6 [Proposed]: Another Filter Step and One EM Step — Task

- **Question:** why does a small positive return point to the calm regime, and why is the EM update a weighted mean?
- **Model:** A5 [Solved]
- **Inputs:** calm $\mu_1 = 0.4$, $\sigma_1 = 2$; turbulent $\mu_2 = -1$, $\sigma_2 = 6$; $p_{11} = 0.95$, $p_{22} = 0.80$; weekly returns in %; last week $P(\text{turbulent}) = 0.1$, this week $r_t = +1\%$
- **Tasks**
 1. Compute the durations, the stationary probabilities and the unconditional mean and standard deviation.
 2. Do one filter step for $r_t = +1\%$.
 3. Compute the probability of the turbulent regime two weeks ahead, $(1 - p, p)P^2$.
 4. Given smoothed probabilities ξ_{tj} , maximise $\sum_t \sum_j \xi_{tj} \ln \varphi(r_t; \mu_j, \sigma_j^2)$ over μ_j and σ_j^2 (the M-step of EM) [Hamilton \(1990\)](#).
- **Report:** the numbers of steps 1–3 and the two M-step formulas
- **Interpretation:** how does each week contribute to the estimate of each regime?
- **Notebook:** section A6

B8 [Proposed]: Bitcoin: Two or Three Regimes? — Task

- ▣ **Question:** does Bitcoin need a third regime, and which regime do its booms belong to?
- ▣ **Model:** B7 [Solved]
- ▣ **Data:** weekly log returns of Bitcoin in %, 2014–2026 (626 weeks)
- ▣ **Tasks**
 1. Estimate two and three regimes with switching mean and variance (30 random starts; check that no regime standard deviation collapses towards 0).
 2. Report the log-likelihood, AIC, BIC and each regime's mean, standard deviation and duration; plot the smoothed probabilities of the three regimes.
 3. Test two vs three regimes with $LR = 2(\hat{\ell}_3 - \hat{\ell}_2)$: simulate 199 samples from the fitted two-regime model, refit both models on each, and compute $p = (1 + \#\{LR^* \geq LR\})/(B + 1)$ [Hansen \(1992\)](#).
 4. Compare the bootstrap 95% value with the χ^2_6 value.
- ▣ **Report:** the table, the chart, LR, the bootstrap p -value and both critical values
- ▣ **Interpretation:** why is χ^2 the wrong reference distribution?
- ▣ **Notebook:** section B8

A7 [Solved]: LPPLS: Slaving the Linear Parameters — Task

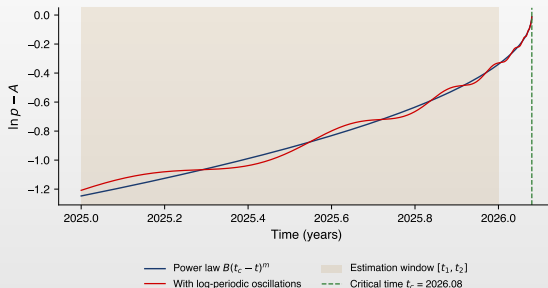
- **Question:** why is LPPLS calibrated over three nonlinear parameters only, and which filter conditions can be checked from the parameters alone?
- **Inputs:** $\ln p_t = A + Bf_t + C_1g_t + C_2h_t + u_t$ on $[t_1, t_2]$, $f_t = (t_c - t)^m$, $g_t = f_t \cos(\omega \ln(t_c - t))$, $h_t = f_t \sin(\omega \ln(t_c - t))$
[Filimonov & Sornette \(2013\)](#)
 - ▶ Filter of Shu and Zhu (eq. 12) [Shu & Zhu \(2020\)](#): $B < 0$; $m \in [0.01, 0.99]$; $\omega \in [2, 25]$; $t_c - t_2 \leq (t_2 - t_1)/5$;
oscillations $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$; damping $D = \frac{m|B|}{\omega|C|} \geq 1$; relative error, Lomb and residual tests (these need data)
- **Tasks**
 1. For fixed (t_c, m, ω) , derive (A, B, C_1, C_2) in closed form.
 2. Write the profile objective left for the nonlinear search.
 3. Recover C and ϕ of $C \cos(\omega \ln(t_c - t) - \phi)$ from C_1, C_2 .
 4. A fit on $[2025.0, 2026.0]$ gives $t_c = 2026.08$, $m = 0.5$, $\omega = 8$, $B = -1.2$, $C_1 = 0.06$, $C_2 = -0.02$: check every condition that the parameters alone allow and plot the implied trend.
- **Report:** the closed form, the profile, C , ϕ and a pass/fail list
- **Interpretation:** why does slaving make the calibration stable?
- Notebook: section A7

A7 [Solved]: Solution (1/2) — Steps 1–3

- Step 1: with $X_t = (1, f_t, g_t, h_t)$ and $\beta = (A, B, C_1, C_2)^\top$, minimise $\sum_t (\ln p_t - X_t \beta)^2$: $\hat{\beta} = (X^\top X)^{-1} X^\top y$, a 4×4 system of sums such as N , $\sum f_t$, $\sum f_t g_t$, $\sum h_t^2$
- Step 2: profile $S(t_c, m, \omega) = y^\top y - y^\top X (X^\top X)^{-1} X^\top y$: only three nonlinear parameters, which removes most local minima of the seven-parameter search
- Step 3: $C \cos(x - \phi) = C \cos \phi \cos x + C \sin \phi \sin x$, so $C_1 = C \cos \phi$, $C_2 = C \sin \phi$: $C = \sqrt{C_1^2 + C_2^2} = 0.0632$, $\phi = \text{atan2}(C_2, C_1)$

A7 [Solved]: Solution (2/2) — Step 4 and the Implied Trend

- Step 4: $B = -1.2 < 0$; $m = 0.5$ and $\omega = 8$ inside their ranges: pass
- $t_c - t_2 = 0.08 \leq 0.2$, i.e. 29 days after the window: pass
- $O = \frac{8}{\pi} \ln \frac{1.08}{0.08} = 6.63 \geq 2.5$;
 $D = \frac{0.5 \cdot 1.2}{8 \cdot 0.0632} = 1.19 \geq 1$: pass
- The relative error, the Lomb test and the residual test need the price data: a candidate, not yet a qualified fit



A8 [Proposed]: Why the Damping Condition? The Hazard Rate — Task

- **Question:** where does the damping condition of the LPPLS filter come from, and does a given fit satisfy it?
- **Model:** A7 [Solved]
- **Inputs:** JLS: with zero interest and no dividends, $\mathbb{E}_t[dp] = 0$ before the crash requires the drift $\mu(t) = \kappa h(t)$, $h(t) \geq 0$ the crash hazard rate and $\kappa > 0$ the crash size; log-price trend $B\tau^m + C\tau^m \cos(\omega \ln \tau - \phi)$, $\tau = t_c - t$ [Johansen, Ledoit & Sornette \(2000\)](#)
- **Tasks**
 1. Differentiate the trend with respect to t .
 2. Find the condition on (B, C, m, ω) for $h(t) \geq 0$ at every phase $\theta = \omega \ln \tau - \phi$.
 3. Show that for $\omega \gg m$ it becomes the damping condition $D = \frac{m|B|}{\omega|C|} \geq 1$ (Shu and Zhu, eq. 11) [Shu & Zhu \(2020\)](#).
 4. For the fit $t_c = 2026.45$, $m = 0.95$, $\omega = 3.5$, $B = -0.8$, $C_1 = 0.25$, $C_2 = 0$ on $[2025.0, 2026.0]$, check the exact condition, D , O and the t_c range.
- **Report:** the three derivations and the four checks
- **Interpretation:** what would a negative hazard rate mean economically, and is the fit a bubble signature?
- **Notebook:** section A8

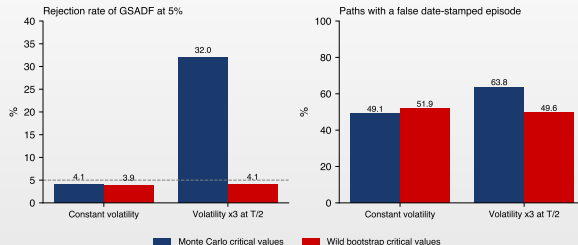
B6 [Proposed]: Does the Confidence Indicator Beat Chance? — Task

- ▣ **Question:** is a positive LPPLS confidence indicator for the Nasdaq 100 followed by large falls more often than no signal?
- ▣ **Model:** B5 and B3 [Solved]
- ▣ **Data:** Nasdaq 100 daily closes; LPPLS confidence indicator from 1997 (Shu and Zhu grid: 141 windows of 750 to 50 trading days, t_2 every 5 trading days) [Shu & Zhu \(2020\)](#)
 - ▶ **Signal:** indicator > 0 ; **outcome:** a close at least 20% below the signal-date close within 90 **calendar** days; dates with an incomplete window are dropped
- ▣ **Tasks**
 1. Find when the indicator was first positive and its share of positive dates between January 1999 and the March 2000 peak.
 2. Estimate $\Delta = \hat{p}_{\text{on}} - \hat{p}_{\text{off}}$ with a 95% CI from a circular block bootstrap (90 days = 13 dates, 9 999 replications) and a Newey–West test of $b = 0$ (13 lags) in a linear probability regression.
 3. Compare the block-bootstrap SE with the independent-data SE and compute the design effect $(\text{SE}_{\text{block}}/\text{SE}_{\text{iid}})^2$.
 4. Compute the sample size for 80% power (equal groups, $p_0 = \hat{p}_{\text{off}}$, $p_1 = p_0 + 10$ points, two-sided 5%), for independent observations and for dependent dates.
- ▣ **Report:** the chart of the indicator, Δ with its CI and p -value, the design effect and the sample sizes
- ▣ **Interpretation:** would the indicator have helped an investor in 2000, and is absence of evidence here evidence of no information?
- ▣ **Notebook:** section B6

B9 [Solved]: Size of GSADF Under a Volatility Shift — Task

- **Question:** how often does GSADF reject a true random walk when volatility triples halfway, and does the wild bootstrap fix it?
- **Model:** B3 [Solved]
- **Data:** simulated; 1 000 random walks $y_t = T^{-1} + y_{t-1} + \sigma_t z_t$, $z_t \sim N(0, 1)$, $T = 400$, $w_0 = 40$; $\sigma_t = 1$ up to $T/2$, then 3: no bubble [Harvey et al. \(2016\)](#)
- **Tasks**
 1. Compute the rejection rate of GSADF at 5% with the Monte Carlo critical value for constant variance (2.19).
 2. Repeat with a wild bootstrap critical value computed for each path (199 replications, as in B3).
 3. Compute the share of paths with at least one false date-stamped episode (BSADF above its pointwise 95% value for $\lceil \ln T \rceil$ periods).
 4. Do the same for constant volatility as a control and plot the four rates.
- **Report:** a table and a bar chart of the rejection and false-episode rates
- **Interpretation:** which critical value would you trust for Bitcoin, and does the bootstrap solve the multiplicity of date-stamping?
- **Notebook:** section B9

B9 [Solved]: Solution (1/2) — Chart and Table



	GSADF, MC	GSADF, wild	false ep., MC	false ep., wild
Constant	4.1%	3.9%	49.1%	51.9%
$\times 3$ at $T/2$	32.0%	4.1%	63.8%	49.6%

B9 [Solved]: Solution (2/2) — Steps and Interpretation

- Steps: simulate the path; compute GSADF and BSADF; compare with the fixed Monte Carlo value and with the path's own wild bootstrap values; average over the 1000 paths (Monte Carlo error ± 1.4 points near 5%)
- The late volatility rise looks like acceleration: 32.0% false rejections with the constant-variance value; the wild bootstrap restores 4.1%
- False episodes in about half of the paths with either value: the bootstrap fixes the size of GSADF, not the multiplicity of date-stamping
- Interpretation: for Bitcoin use the wild bootstrap for the global test and family-wise critical values for the dates

B10 [Proposed]: Size Under GARCH Volatility — Task

- **Question:** does persistent GARCH volatility, or a fall in volatility, distort the size of GSADF?
- **Model:** B9 [Solved]
- **Data:** simulated, as in B9, with two new null models: GARCH(1,1) shocks $\sigma_t^2 = 0.02 + 0.10 e_{t-1}^2 + 0.88 \sigma_{t-1}^2$ (unconditional variance 1), and a volatility **fall** from 3 to 1 at $T/2$
- **Tasks**
 1. Compute the size of GSADF at 5% with Monte Carlo and with wild bootstrap critical values (1 000 paths, 199 replications each).
 2. Compute the share of paths with a false date-stamped episode and plot the rates as in B9.
- **Report:** a table and a bar chart as in B9
- **Interpretation:** why does a fall in volatility not create false bubbles while a rise does, and is the wild bootstrap exact under GARCH?
- **Notebook:** section B10

B2 [Proposed]: SADF vs GSADF — Task

- ▣ **Question:** which episodes does the recursive ADF (fixed start) miss that BSADF (moving start) finds?
- ▣ **Model:** B1 [Solved]
- ▣ **Data:** monthly log closes of the Nasdaq 100 and of the S&P 500, 1990–2026 (last close of each month)
- ▣ **Tasks**
 1. Compute SADF and GSADF with the minimum-window rule and their 95% critical values from 2000 simulated random walks.
 2. Date-stamp with the recursive ADF of PWY (window start fixed at the first month) and with BSADF of PSY, minimum duration $\lceil \ln T \rceil$ months [Phillips, Wu & Yu \(2011\)](#).
 3. List the episodes of both methods with their direction and plot both statistics with their critical values.
- ▣ **Report:** a table of SADF and GSADF, both lists of episodes and the chart
- ▣ **Interpretation:** why does the fixed start miss the later episodes?
- ▣ **Notebook:** section B2

C1 [Proposed]: Is There an AI Bubble? — Task

- **Question:** is the current rise of NVIDIA and the Nasdaq 100 more like the dot-com bubble of Cisco, or like explosive fundamentals?
- **Model:** B1, B3, B5 [Solved]; B6
- **Data:** monthly and weekly closes of NVIDIA, Cisco and the Nasdaq 100; quarterly sales and earnings from the companies' filings (SEC EDGAR, 10-Q/10-K)
- **Tasks**
 1. Run GSADF and BSADF on monthly and weekly data with Monte Carlo and wild bootstrap critical values; date the signals and say whether the current episode is still open.
 2. Compute two-year run-ups, raw and net of the Nasdaq 100, as in Greenwood, Shleifer and You [Greenwood, Shleifer & You \(2019\)](#).
 3. Compute the LPPLS confidence indicator over the last 26 weeks.
 4. Build the fundamentals yourselves: filings dated at the filing date (no look-ahead), trailing four-quarter sums, split-consistent per-share values, sales when earnings are ≤ 0 .
 5. Test the log price-to-sales (or price-to-earnings) ratio and date-stamp it with family-wise critical values [Phillips & Shi \(2020\)](#).
- **Report:** a two-page note with charts and a Quantlet folder (code, Metainfo, charts)
- **Interpretation:** can explosive prices be separated from explosive expected profits [Pástor & Veronesi \(2009\)](#)?

C3 [Proposed]: Does Volume Lead Prices in a Speculative Boom? — Task

- **Question:** does trading volume lead prices in a speculative boom, as in the US housing cycle of 2000–2011 [DeFusco, Nathanson & Zwick \(2022\)](#)?
- **Model:** B6 (block bootstrap), B3 (Bitcoin episodes); the lecture case study
- **Data:** [FRED](#): monthly Case-Shiller U.S. National Home Price Index (CSUSHPINSA) and new one-family houses sold, not seasonally adjusted (HSN1FNSA), 2000–2011 and 1990–2011
 - ▶ The paper regresses the demeaned log house price on sales lagged k months (Eq. 1) in 115 cities; the correlation peaks at $k = 24$ (Figure 2)
- **Tasks**
 1. Build p_t , the log price minus its sample mean, and v_t , sales minus the mean of their calendar month.
 2. For $k = -12, \dots, 48$ estimate Eq. (1) by OLS without a constant, compute the implied correlation $\hat{\beta}_k \text{sd}(v_{t-k})/\text{sd}(p_t)$ and plot it against k .
 3. Give pointwise 95% CIs at the peak lag, at $k = 24$ and at $k = 0$ from a circular block bootstrap with 24-month blocks; repeat for 1990–2011.
 4. Date the quiet: from the peak of adjusted volume to the last peak of prices before their decline (Figure 1).
 5. Extension: repeat steps 1–4 for Bitcoin (BTC-USD.CC, month-end close, monthly summed volume) around a BSADF episode (B3); justify any seasonal adjustment [Griffin & Shams \(2020\)](#).
- **Report:** the correlation curve, the peak lag with its CI, the quiet dates, a Quantlet folder and a two-page note
- **Interpretation:** new-home sales exclude resales by short-term buyers: does a national lead of volume confirm speculation, or could credit or construction cycles produce it?

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